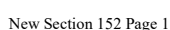
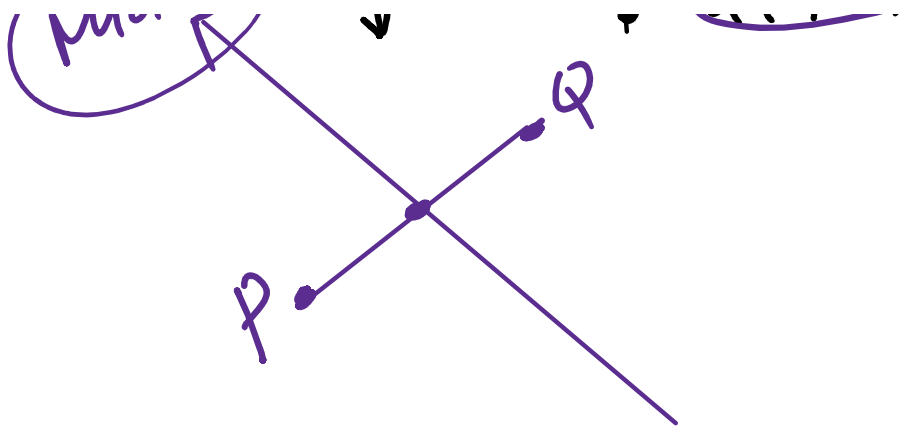


Chapter 07 Coordinate Geometry

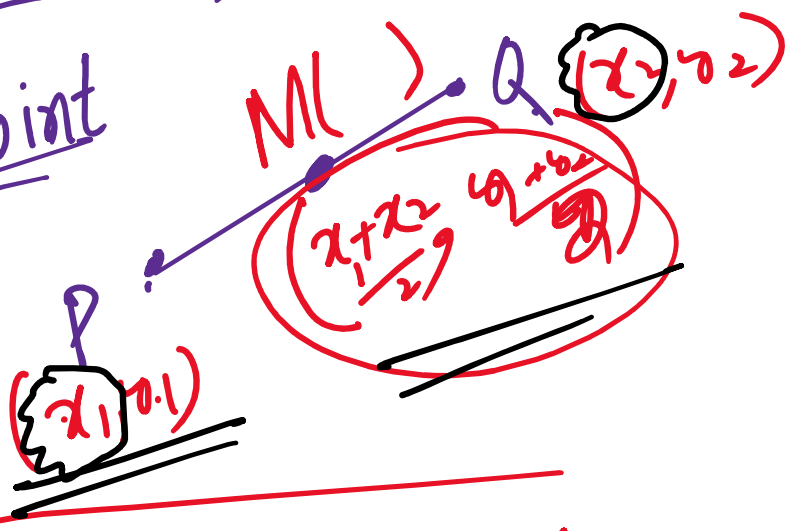
Mid Point , Distance Formula





$P(x_1, y_1)$, $Q(x_2, y_2)$

Mid-Point



Distance Formula.

Always +ve

$$\begin{aligned} 2, 3 \\ 2-3 &= (-1)^2 = 1 \\ 3-2 &= (1)^2 = 1 \end{aligned}$$

d $P(x_1, y_1)$ $Q(x_2, y_2)$

$$\checkmark d = \sqrt{(x_1 - x_2)^2 + (y_2 - y_1)^2}$$

$$\checkmark d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\checkmark d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

1. Encircle the correct option in the following.

$$AB = 2.95, CD = 3.20$$

(i) If the coordinates of four points are $A\left(\frac{3}{2}, -\frac{7}{4}\right)$, $B\left(3, \frac{3}{4}\right)$, $C(3, -2)$ and $D\left(1, \frac{1}{2}\right)$, the relationship between the distance AB and CD is:

- (a) $AB > CD$ (b) $AB < CD$ (c) $AB = CD$ (d) $AB = \frac{1}{2} CD$

(ii) If a and b are any two points on a coordinate line (number line), then its mid point will be:

- (a) $\frac{a+b}{2}$ (b) $\left(\frac{a}{2}, \frac{b}{2}\right)$ (c) $\left(\frac{a}{2}, b\right)$ (d) $\left(a, \frac{b}{2}\right)$

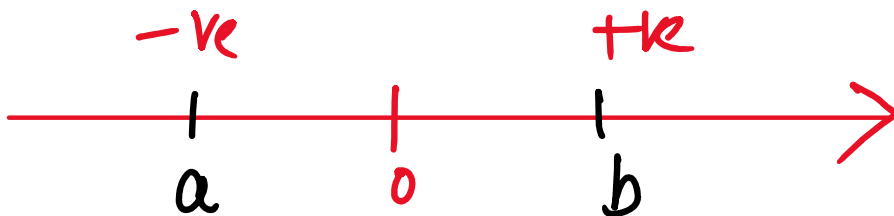
(iii) (5, 2), (7, 0), (1, -2) and (-1, 0) are the vertices of

$$AB = \sqrt{\left(3 - \frac{3}{2}\right)^2 + \left(\frac{3}{4} + \frac{7}{4}\right)^2}$$

(ii) If a and b are any two points on a coordinate line (number line), then its mid point will be:

- (a) $\frac{a+b}{2}$ (b) $\left(\frac{a}{2}, \frac{b}{2}\right)$ (c) $\left(\frac{a}{2}, b\right)$ (d) $\left(a, \frac{b}{2}\right)$

(iii) (5, 2), (7, 0), (1, -2) and (-1, 0) are the vertices of



$$\frac{a+b}{2} \rightarrow \text{Mid Pon}$$

3, 4, 5

Mean, Average

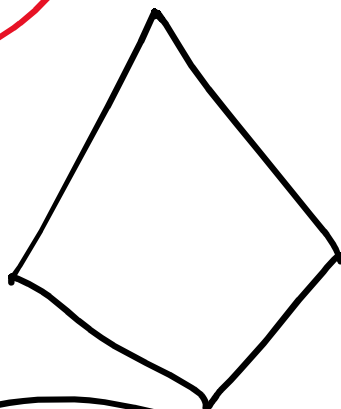
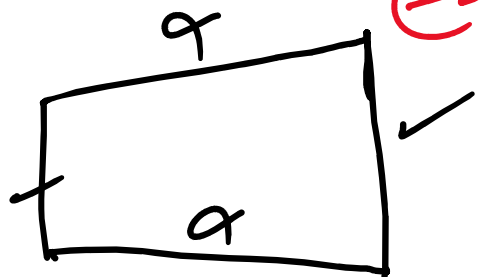
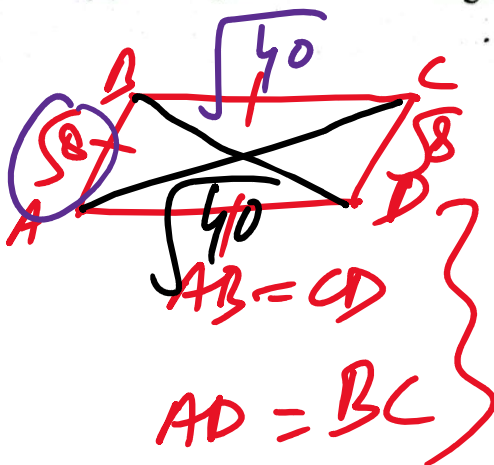
Mid Point

$$d_1 = \sqrt{4+4} = \sqrt{8} = 2\sqrt{2}$$

$$d_2 = \sqrt{4+4} = \sqrt{8} = 2\sqrt{2}$$

$$\frac{3+5}{2} = \frac{8}{2} = 4$$

- (iii) (5, 2), (7, 0), (1, -2) and (-1, 0) are the vertices of a
- (a) parallelogram (b) square (c) trapezium (d) kite ✓
- (iv) The triangle with vertices (1, -3), (3, 2) and (-2, 4) is:
- (a) a right triangle (b) an isosceles triangle ✓
- (c) a right isosceles triangle (d) an equilateral triangle ✓



$$(5, 2), (-1, 0)$$

$$\sqrt{(5 - (-1))^2 + 2^2}$$

$$\sqrt{(5-(-1))^2 + 4}$$

$$\sqrt{36+4} = \sqrt{40}$$

$0-(-2)$

$$(7, 0), (1, -2)$$

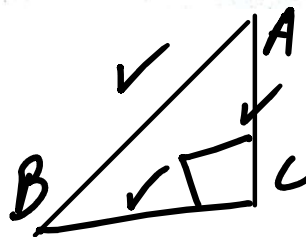
$$\sqrt{36+4} = \sqrt{40}$$

(iii) ^A(5, 2), ^B(7, 0), ^C(1, -2) and ^D(-1, 0) are the vertices of a ^(2, -) ^(u, 2)

(iv) ^(a) parallelogram ^(b) square ^(c) trapezium ^(d) kite

The triangle with vertices (1, -3), (3, 2) and (-2, 4) is:

^(a) a right triangle ^(b) an isosceles triangle ^(c) a right isosceles triangle ^(d) an equilateral triangle



(iv)

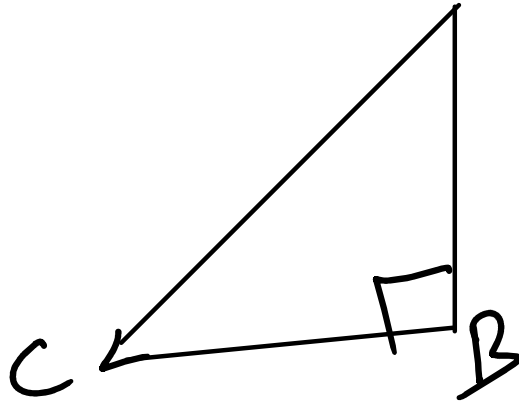
$$AB = \sqrt{4+25} = \sqrt{29} \checkmark$$

$$BC = \sqrt{25+4} = \sqrt{29} \checkmark$$

$$AC = \sqrt{9+49} = \sqrt{58}$$

$AC^2 = 58$, $BC^2 = 28$
 $AB^2 = 28$

$$AC^2 = AB^2 + BC^2$$



(v) Mid-point of $(c, -d)$ and $(-d, c)$ is:

(a) $\left(\frac{c}{2}, -\frac{d}{2}\right)$

(b) $\left(\frac{-c-d}{2}, \frac{-d-c}{2}\right)$

(c) $\left(\frac{-d+c}{2}, \frac{c-d}{2}\right)$

(d) $(c-d, -d+c)$

$$\frac{c + (-d)}{2} = \frac{c-d}{2} = \frac{-d+c}{2}$$

(vi) If $P(2, 3)$ is the mid point of A $(x, 4)$ and B $(-3, y)$, then the coordinates (x, y) are:

(a) $\left(\frac{-1}{2}, \frac{3+b}{2}\right)$ (b) $(5, -1)$

(c) $\left(\frac{x-3}{2}, \frac{y+4}{2}\right)$

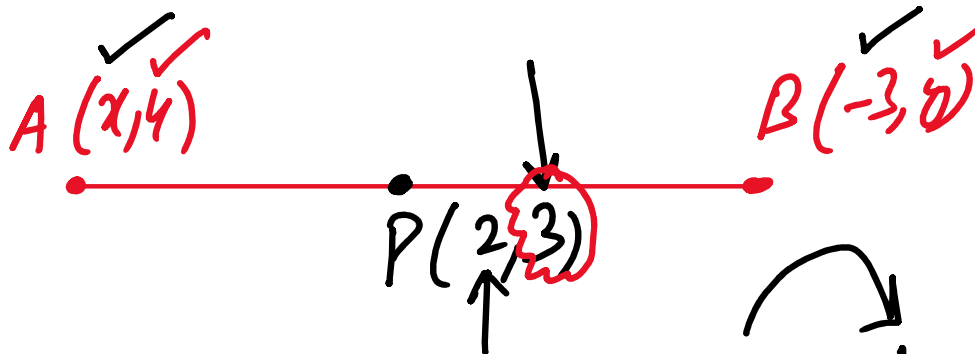
(d) $(7, 2)$

- (vi) ✓ If $P(2, 3)$ is the mid point of A $(x, 4)$ and B $(-3, y)$, then the coordinates (x, y) are:

(a) $\left(\frac{-1}{2}, \frac{3+b}{2}\right)$ (b) $(5, -1)$

(c) $\left(\frac{x-3}{2}, \frac{y+4}{2}\right)$

(d) $(7, 2)$



$$2 = \frac{x-3}{2} \Rightarrow x-3=4$$

$$x = 3+4 = 7$$

$$3 = \frac{4+y}{2} \Rightarrow y+4=6 \Rightarrow y=6-4$$

$$y = 2$$

$P(x, y)$

Abcissa

Ordinate

- (vii) If two ordered pairs $(4, 5)$ and $\left(\frac{a+1}{2}, b-3\right)$, are equal then, the value of a is:

(a) 4

(b) 8

(c) 7 ✓

(d) 5

- (viii) In which quadrant, the point $(-4, -(-6))$ lies?

(a) first

(b) second

(c) third

(d) fourth

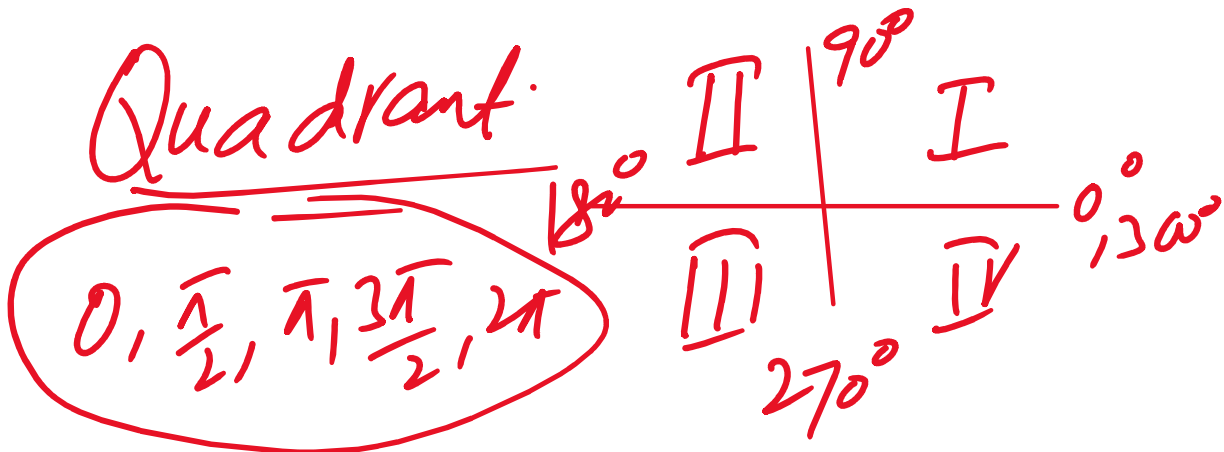
$$\frac{a+1}{2} = 4 \Rightarrow a+1 = 8$$

$$a = 8 - 1 = 7$$

$$\frac{a+b}{2} = \frac{4+12}{2} = 8$$

$$a = 8 - 1 = 7$$

$$b - 3 = 5 \Rightarrow b = 8$$



$\sin$ $\csc$	<u>II</u>	S ^{+ve}	A I All
$\tan$ $\cot$	<u>III</u>	T	C <u>IV</u> $\cos$ ^{+ve} $\sec$ ^{+ve}

Add sugar to coffee

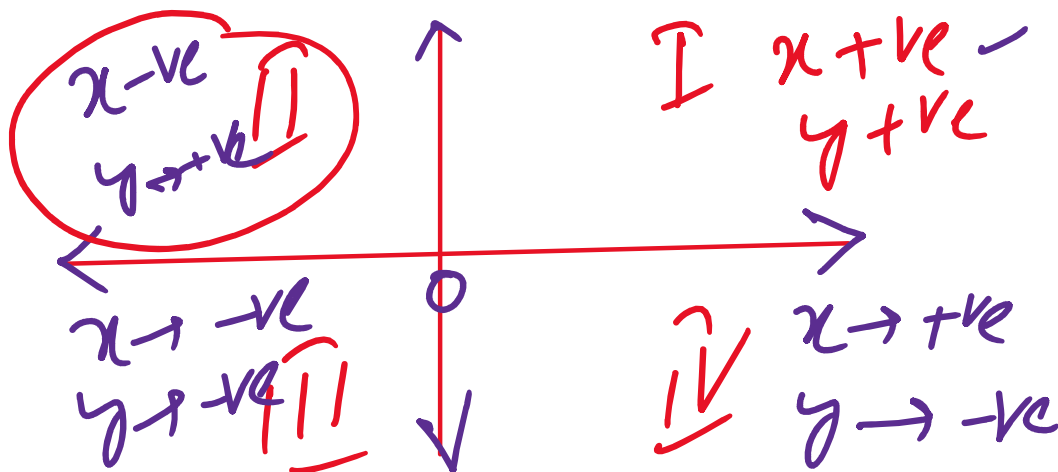
$$\sin \theta < 0 \quad (-ve)$$

$$\cos \theta > 0 \quad (+ve)$$

IVth Que

$$\left. \begin{array}{l} \tan \theta < 0 \\ \sin \theta < 0 \end{array} \right\} \text{---} \underline{\underline{\text{IV}}}$$

$P(x, y)$ $\sin \theta < 0$, $\tan \theta > 0$
3rd Qu



(vii) If two ordered pairs $(4, 5)$ and $\left(\frac{a+1}{2}, b-3\right)$, are equal then, the value of a is:

(a) 4

(b) 8

(c) 7

(d) 5

(viii) In which quadrant, the point $(-4, -6)$ lies?

(a) first

(b) second

(c) third

(d) fourth

$x \rightarrow -ve$
 $y \rightarrow +ve$



(ix) In the adjoining figure of a line:

$d(A, C) - d(B, C) =$

(a) $d(B, A)$

(b) $d(C, B)$

(c) $d(A, B)$

(d) both a & c

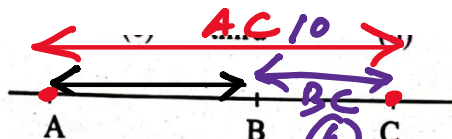
(x) The abscissa on y -axis is:

(a) 1

(b) x

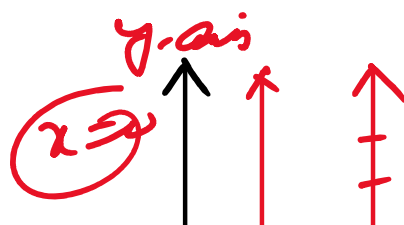
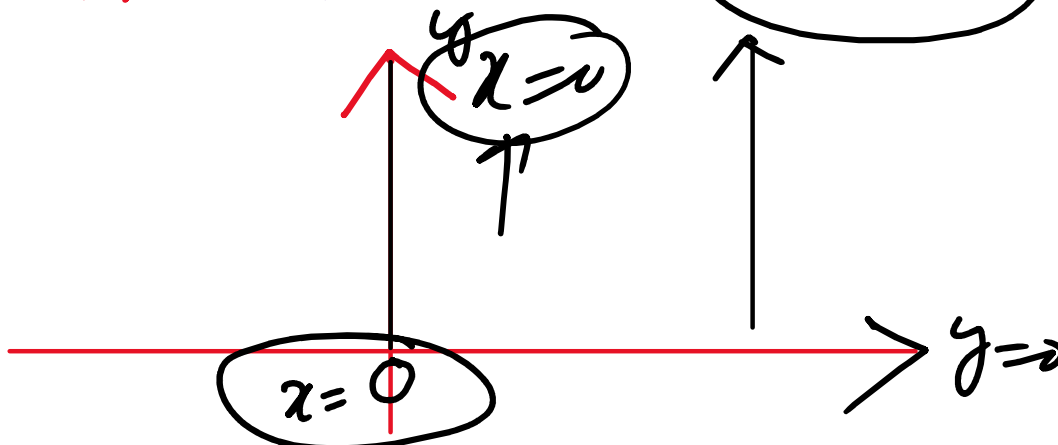
(c) zero

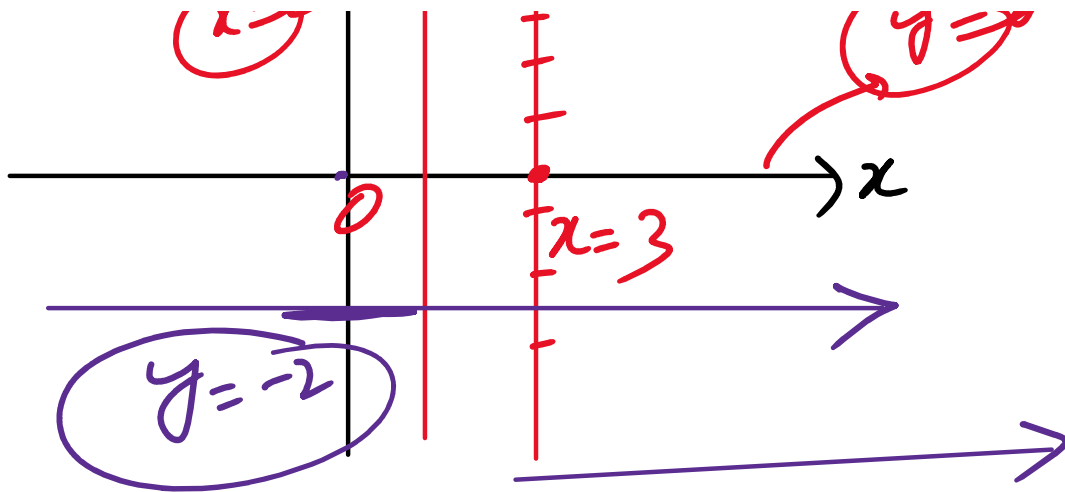
(d) y



x -coordinate.
 $AC - BC$

$AB = BA$





(xi) Minimum number of sides in a polygon is:

(a) 2

(b) 3

(c) 4

(d) 5

(xii) If $x < 0$ and $y > 0$, then $(-x, y)$ will lie in quadrant:

(a) I

(b) II

(c) III

(d) IV

(xiii) The point $(3, 5)$ is at minimum distance from:

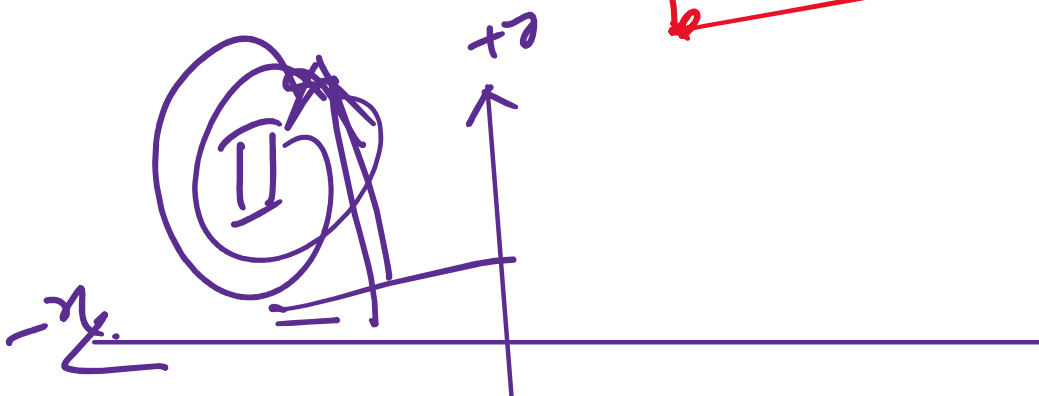
(a) x-axis

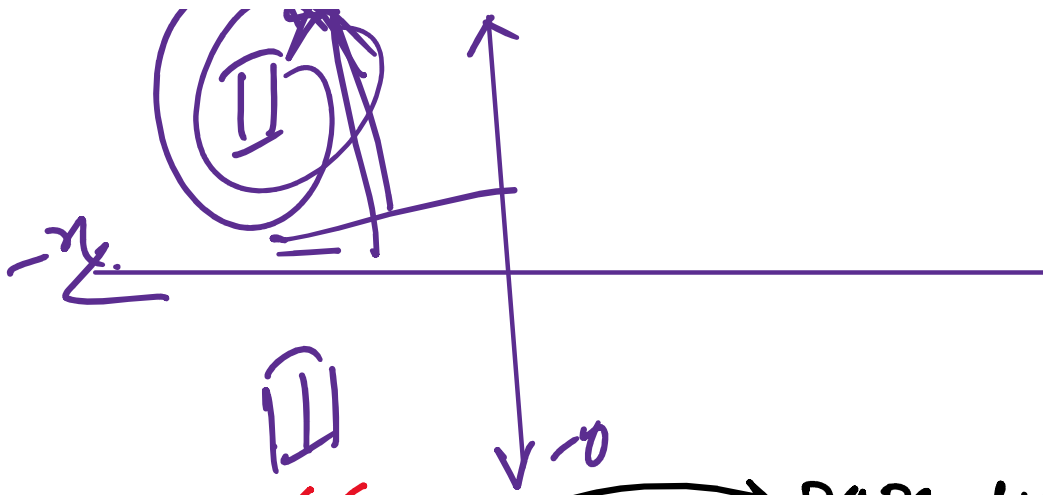
(b) y-axis

(c) origin

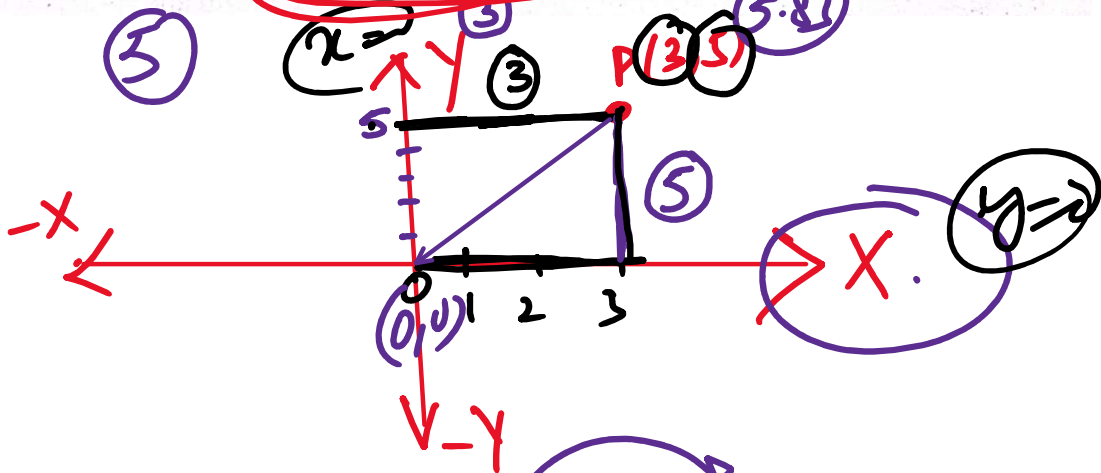
(d) both a & b

If ^{are} more than 2 sides $\Rightarrow$ polygon
no. of sides





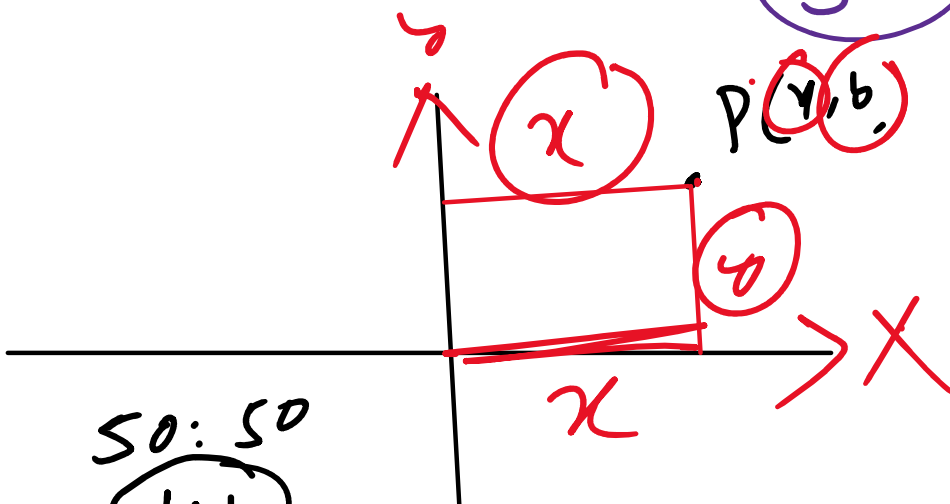
- (xiii) The point (3, 5) is at minimum distance from:
- (a) x-axis (b) y-axis (c) origin (d) both a & b



$$(3, 5), (0, 0)$$

$$\sqrt{3^2 + 5^2} = \sqrt{9 + 25}$$

$$= \sqrt{34}$$



50:50
1:1

(xiv) If M is the mid point of the line segment LN, then LN : MN is:

(a) 2:1

(b) 1:2

(c) 1:1

(d) 1:3

(xv) All the bisectors of a line segment always pass through:

(a) x-axis

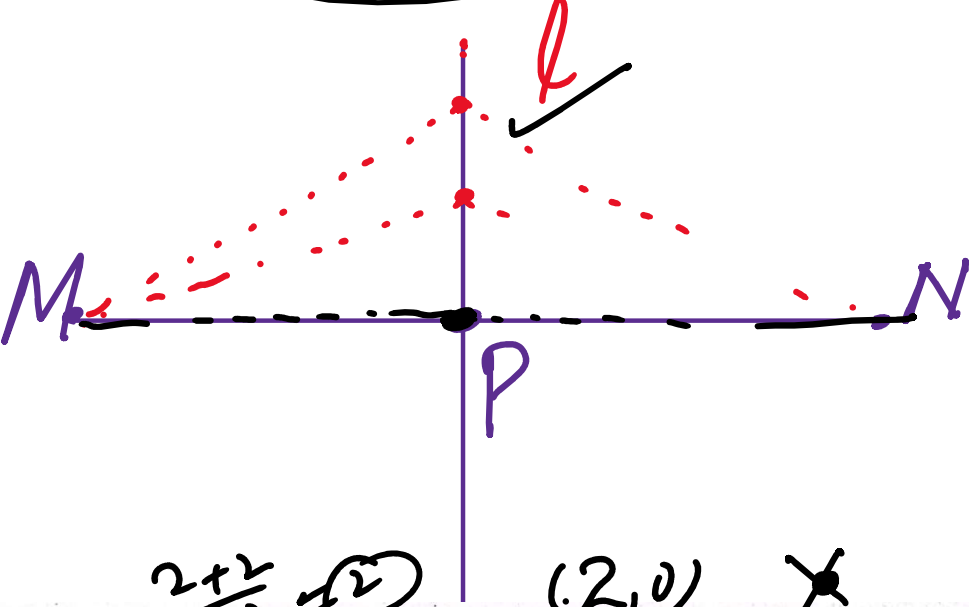
(b) origin

(c) mid-point

(d) y-axis



LN : MN
2 : 1



$2 + \frac{2}{2} = 2$

(2, 0)

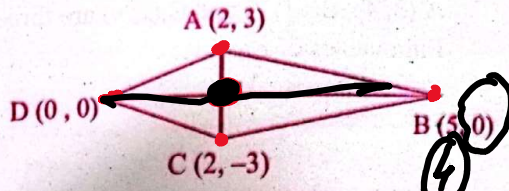
(xvi) The below figure is a kite, what will be the coordinates of the point of intersection of both diagonals?

(a) (0, 2)

(b) $(2\frac{1}{2}, 0)$

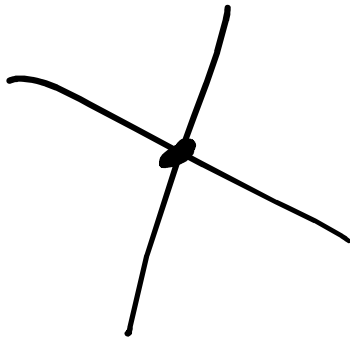
(c) (0, 0)

(d) (2, 0)



$\frac{3+3}{2} = 3$
 $(0, 0), (5, 0) \Rightarrow$

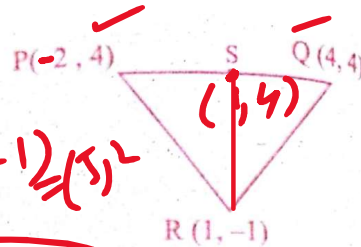
$\frac{5+0}{2} = \frac{5}{2}$



(xvii) In the given figure, if S is the mid point of PQ, then SR is

- (a) $\sqrt{2}$
(c) 6

- (b) ± 5
(d) 5



Handwritten calculations for problem (xvii):
 $\frac{4+4}{2} = 4$
 $\frac{4-2}{2} = 1$
 $S(1, 4), R(1, -1)$
 (-5)

Handwritten calculation for distance SR:
 $d = \sqrt{0^2 + 25} = 5$

(xviii) If distance between two points A(c, 10) and B(0, -1) is 20 units, then the value of c is:

- (a) 279

- (b) $3\sqrt{31}$

- (c) 3

- (d) 40

Handwritten equation for problem (xviii):
 $d = \sqrt{c^2 + 121} = 20$

Handwritten equation for problem (xviii):
 $c^2 + 121 = 400$

Handwritten equation for problem (xviii):
 $\sqrt{c^2} = \sqrt{400 - 121}$

Handwritten result for problem (xviii):
 $c = 17$

(xix) The length of the hypotenuse of $\triangle XYZ$ whose vertices are (7, 4), (7, 1) and (-3, 1) is:

- (a) 3

- (b) 10

- (c) $\sqrt{109}$

- (d) $\sqrt{91}$

Handwritten calculations for problem (xix):
 $7 - (-3) = 10$
 $4 - 1 = 3$
 $\sqrt{10^2 + 3^2} = \sqrt{109}$

$$XY = \sqrt{0+9} = 3$$

$$YZ = \sqrt{100+0} = 10$$

$$XZ = \sqrt{100+9} = \sqrt{109} \checkmark$$

$$7 - (-3)$$

$$7 - (-3)$$

$$4 - 1$$

$$\sqrt{4+1} \neq \sqrt{5}$$

(xx) The mid point of the hypotenuse of a right triangle with vertices (4, 0), (2, 1) and (-1, -5) is

(a) $(2, \frac{1}{2})$

(b) $(\frac{1}{2}, -2)$

(c) $(\frac{3}{2}, -\frac{5}{2})$

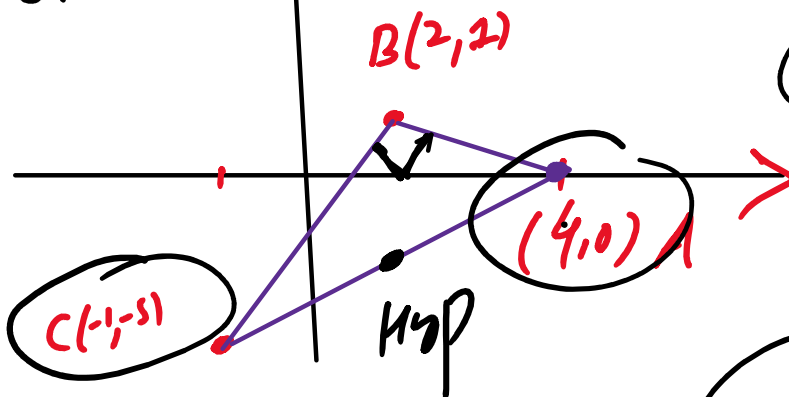
(d) $(\frac{3}{2}, \frac{5}{2})$

~~AD~~

$$\sqrt{9+36} = \sqrt{45}$$

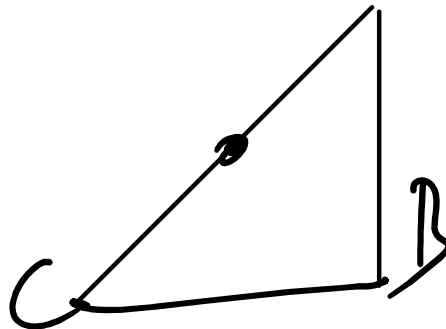
$$\sqrt{25+25}$$

$$\sqrt{50}$$



$$\frac{4-1}{2}, \frac{-5+0}{2}$$

$$(\frac{3}{2}, -\frac{5}{2})$$



(xxi) If end points of a diameter of the circle are (4, 5) and (6, 9), then coordinates of its center will be:

(a) (7, 5)

(b) (1, 2)

(c) (5, 7)

(d) (0, 0)

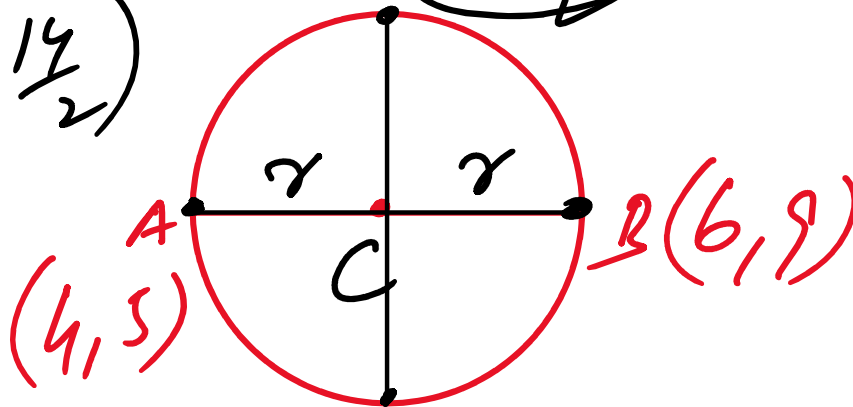
Red

(xxi) If end points of a diameter of the circle are (4, 5) and (6, 9), then coordinates of its center will be:

- (a) (7, 5) (b) (1, 2) (c) (5, 7) (d) (0, 0)

Red

$$\left(\frac{10}{2}, \frac{14}{2}\right)$$



(xxii) If two points (2, 3) and (4, 5) are at equal distance from (3, t) then, the value of t is:

- (a) 5 (b) 4 (c) 3 (d) 7



$$AM = BM$$

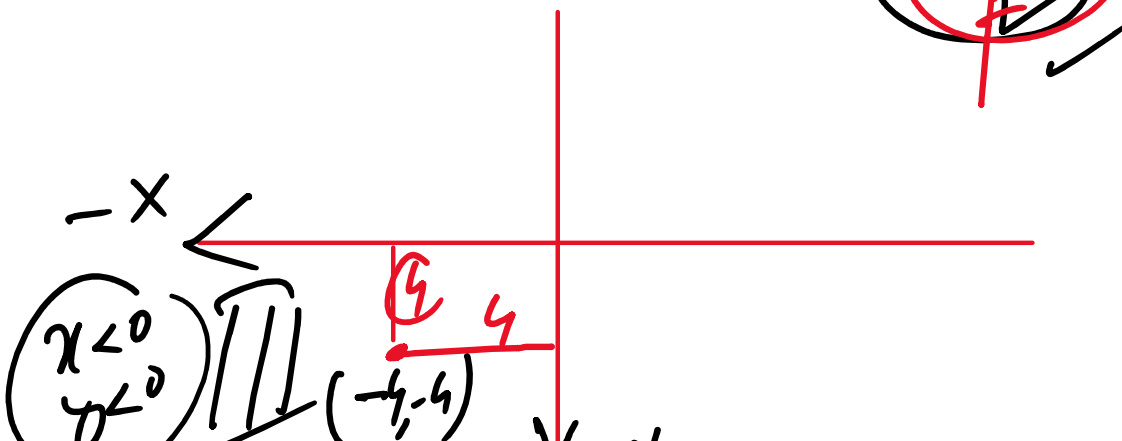
$$x = 4 \checkmark$$

$$\frac{3+5}{2} = t$$

$$t = \frac{8}{2} = 4$$

(xxiii) A point P(x, y) is at equal distance from both the axes and lies in the third quadrant. If it is at a distance of 4 units from y-axis, then its coordinates are:

- (a) (-4, 4) (b) (4, -4) (c) (4, 4) (d) (-4, -4)



$(x, y) \parallel \parallel (-4, -4)$
 $\downarrow \quad \downarrow$
 $x - y$
 Short -

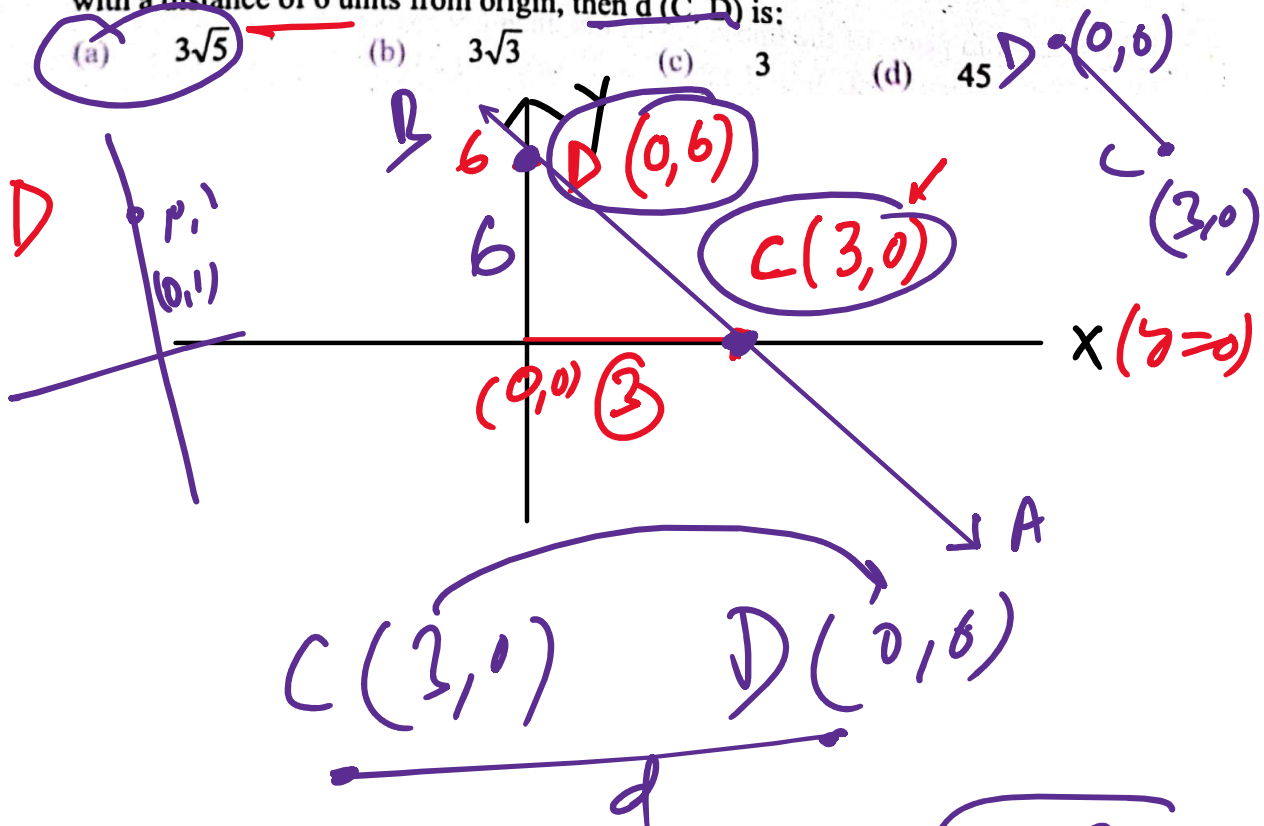
(xxiv) A line AB intersects x-axis at C with a distance of 3 units from origin and y-axis at D with a distance of 6 units from origin, then d(C, D) is:

(a) $3\sqrt{5}$

(b) $3\sqrt{3}$

(c) 3

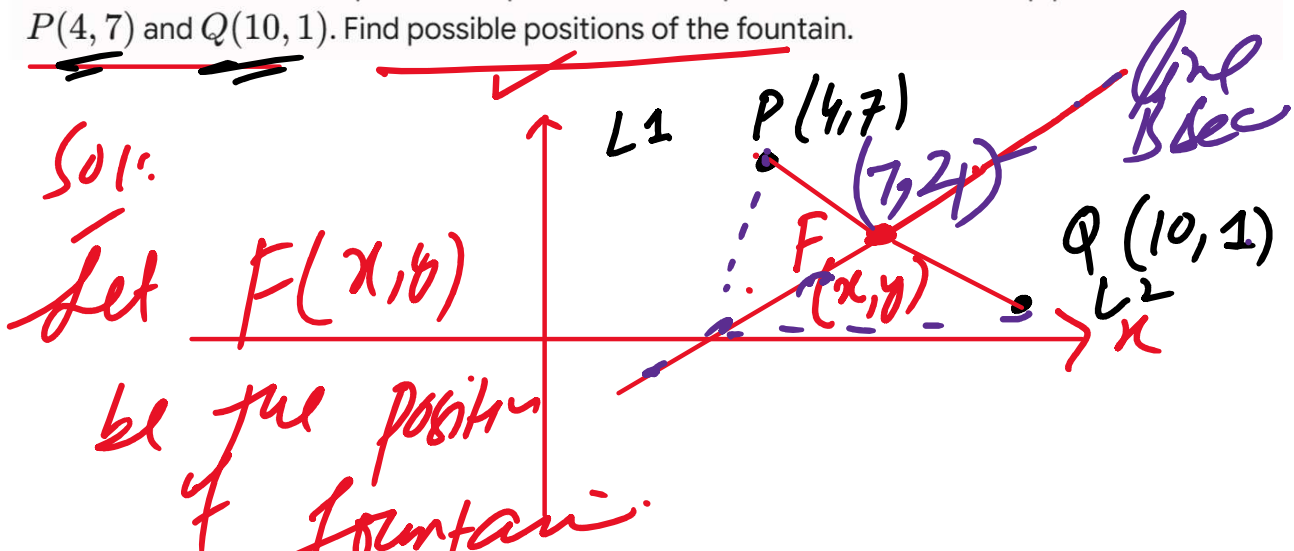
(d) 45



$$d = \sqrt{3^2 + 6^2} = \sqrt{9 + 36} = \sqrt{45}$$

Short Ques 2026

A water fountain is to be placed at a point which is equidistant from two lamp posts located at $P(4, 7)$ and $Q(10, 1)$. Find possible positions of the fountain.



4 fontaine.

Given $|FP| = |FQ|$

$$\sqrt{(x-4)^2 + (y-7)^2} = \sqrt{(x-10)^2 + (y-1)^2}$$

Squaring both sides, —

$$(x-4)^2 + (y-7)^2 = (x-10)^2 + (y-1)^2$$

$$\cancel{x^2} + 16 - 8x + \cancel{y^2} + 49 - 14y = \cancel{x^2} + 100 - 20x + \cancel{y^2} + 1 - 2y$$

$$\begin{array}{r} -8x - 14y + 49 + 20x + y + 16 - 100 = 0 \\ \hline 12x - 12y - 36 = 0 \end{array}$$

$$12x - 12y - 36 = 0$$

$$\div 12$$

$$\boxed{x - y - 3 = 0}$$

$$\boxed{y = x - 3}$$

we show all the regular
possible parking spots

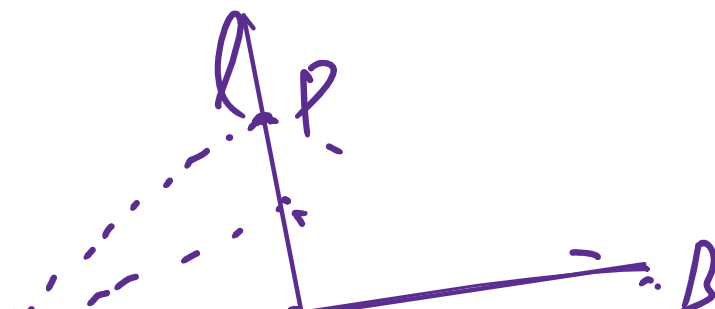
$$y = x - 3$$

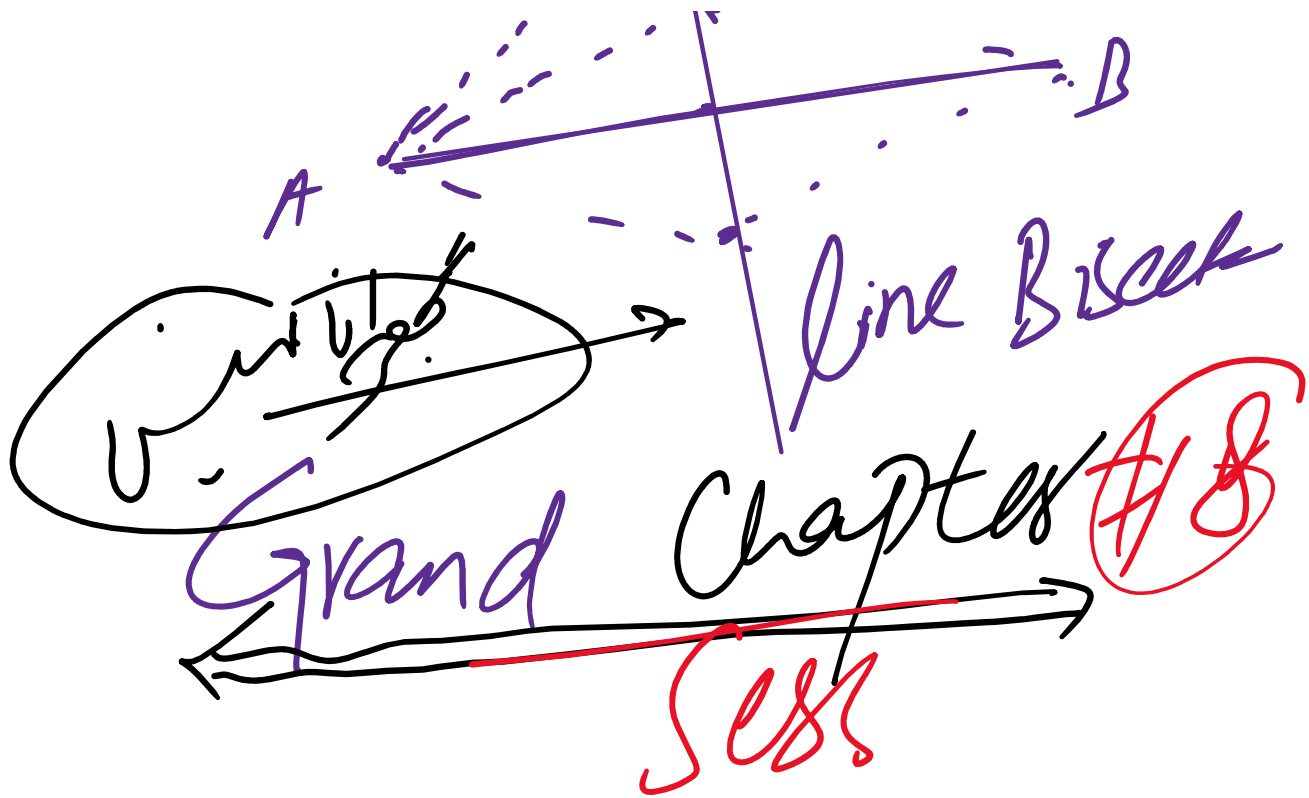
for $x=7$, $y=4$ $P(7,4)$

" $x=8$, $y=5$ $Q(8,5)$

" $x=9$, $y=6$ $R(9,6)$

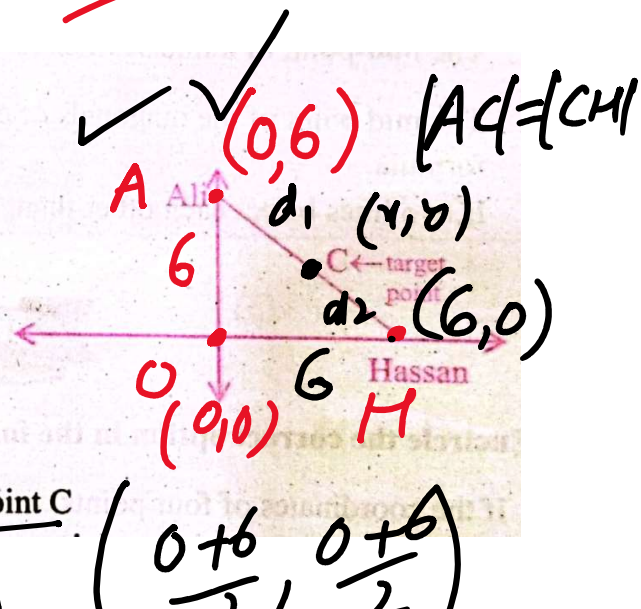
We have infinite
many





6. Hassan and Ali participated in a race competition. They both have to start running from different points on coordinate axes at a distance of 6 units from origin but they must reach at the same target point C, after covering an equal distance. If Hassan starting point is on x-axis and Ali's starting point is on y-axis as shown in the figure, then find the coordinates of the:

- (i) starting points of Hassan and Ali (ii) target point C



$$M(6,0) \quad A(0,6) \quad \left(\frac{0+6}{2}, \frac{0+0}{2} \right) = (3,3) \quad \Delta$$

$$|CA|^2 = |CM|^2$$

$$(x-0)^2 + (y-6)^2 = (x-6)^2 + (y-0)^2$$

$$\cancel{x^2} + \cancel{y^2} + \cancel{36} - 12y = \cancel{x^2} + \cancel{36} - 12x + \cancel{y^2}$$

$$12x - 12y = 0$$

$$x - y = 0$$

$$\textcircled{x = y}$$